

MOTIVATION

Why does this project matter?

Spacecraft engineers today monitor ESA satellite subsystems manually or with simple threshold alarms and a round-the-clock effort across 76+ sensor channels spanning power, thermal, attitude control, and communications. This approach is error-prone and demands large engineering teams.

The open problem:

Kotowski et al. (2024) showed that **existing unsupervised methods consistently produce false alarms** on the ESA-ADB benchmark. But no work asked: *what latent structure actually underlies the telemetry?*

RESEARCH QUESTIONS

Q1: Does ESA satellite telemetry contain consistent latent structure aligned with known operational subsystems?

Q2: Can learning that latent structure improve unsupervised anomaly detection beyond existing threshold-based approaches?

DATASET

ESA Anomaly Dataset

Kotowski et al. 2024 | De Canio et al. 2024

Mission 1: 76 channels | 7.3M train timesteps @ 0.033 Hz | 4 subsystems | 18 groups

Mission 2: 100 channels | 3.1M train timesteps @ 0.056 Hz | 5 subsystems | 29 groups

Labels: Point-wise ESA engineer anomaly annotations per channel

Preprocessing: ESA-ADB scripts; irregular timestamps resampled to uniform Hz

METHOD 1: K-MEANS (RAW FEATURES)

K-Means applied directly to the full StandardScaled 76-channel telemetry matrix. Anomaly score = Euclidean distance from each timestep to nearest K-means centroid. K selected via silhouette score and validation PR-AUC sweep (K = 2–10). Evaluated point-wise vs. ESA ground-truth labels: PR-AUC, ROC-AUC, timeline.

K-MEANS RESULTS

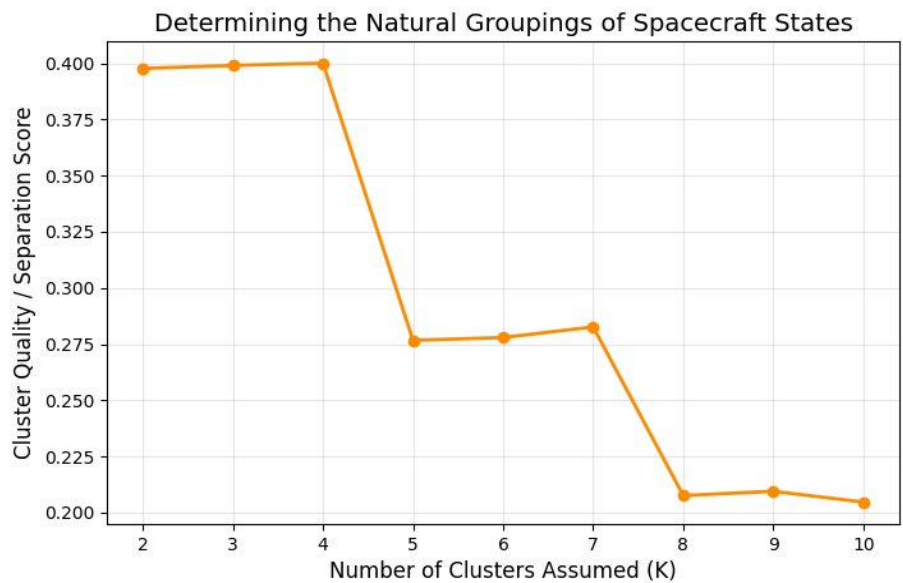


Fig 1. peaks at K=4, consistent with 4 known subsystems; collapses at K=5 subsystems. *Mission 1 only

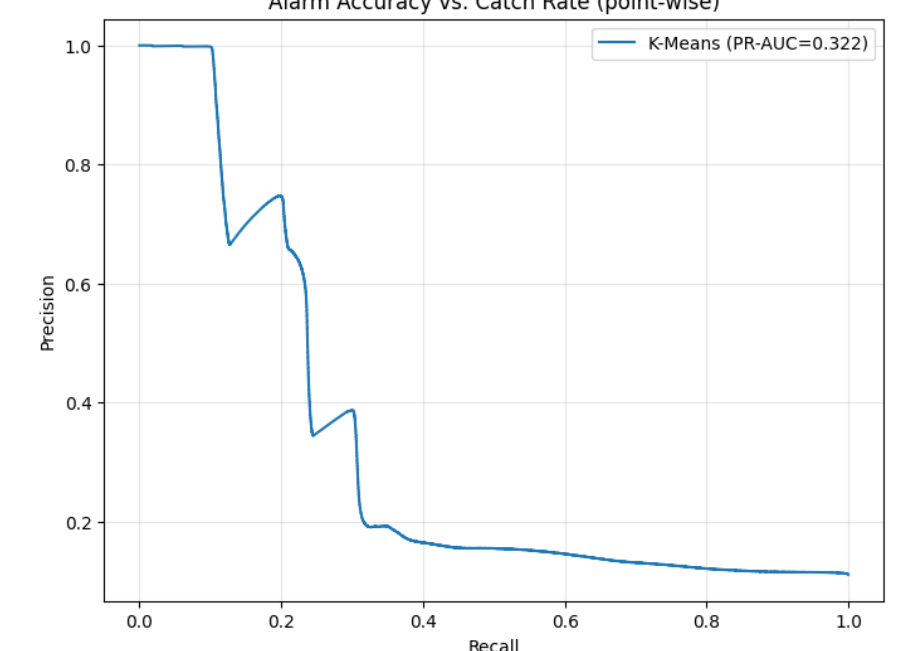


Fig 2. K-Means PR curve (PR-AUC = 0.322). Precision drops sharply at recall > 0.30; anomalies are buried in normal-data mass.

*Mission 1 only

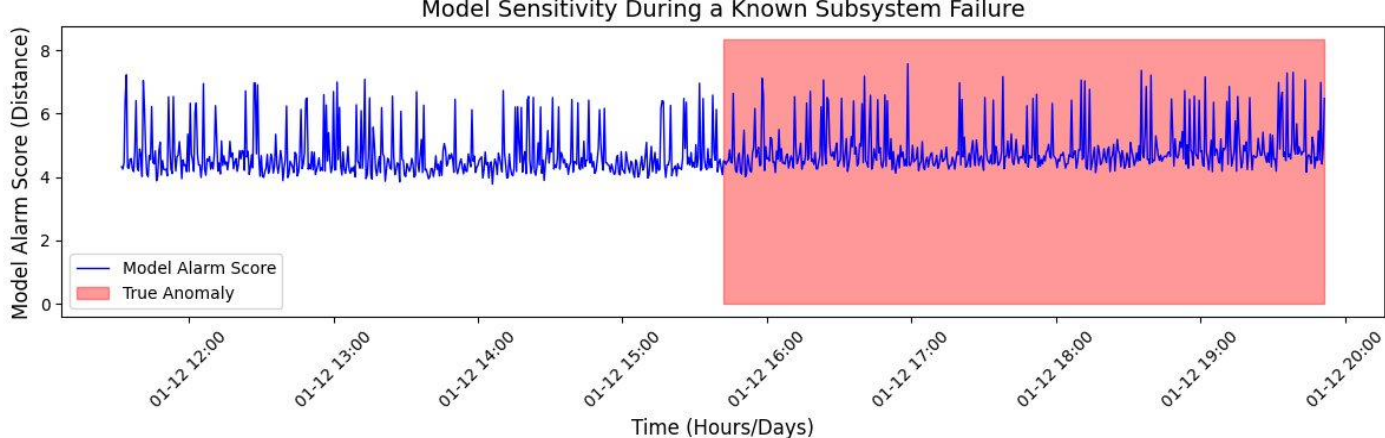


Fig 3. Alarm score (blue) over a known subsystem failure (pink). Score is indistinguishable before vs. during the anomaly — alarm fires equally on normal data. *Mission 1 only

Baseline floor: **PR-AUC = 0.322 | ROC-AUC = 0.634 | K* = 2 (not 4) silhouette peak K=4; anomaly detection optimal K=2**

METHOD 2: PCA + K-MEANS

PCA applied to compress 76 channels to d = 33 (95% variance retained). Two anomaly scores: (1) PCA Reconstruction Error $\|x - \hat{x}\|$ and (2) K-Means centroid distance in PCA space. PC1 feature loadings extracted to test whether unsupervised PCA recovers subsystem-coherent patterns without labels.

RELATED WORK

Kotowski et al. (2024): ESA-ADB benchmark — first large-scale real satellite telemetry dataset with curated anomaly labels; established that existing unsupervised methods produce persistent false alarms. arXiv:2406.17826

Hundman et al. (2018): LSTM + non-parametric dynamic thresholding for spacecraft anomaly detection. ACM KDD 2018.

Lu et al. (2025): Self-supervised autoencoder with multi-scale contrastive learning for multivariate anomaly detection.

PCA RESULTS

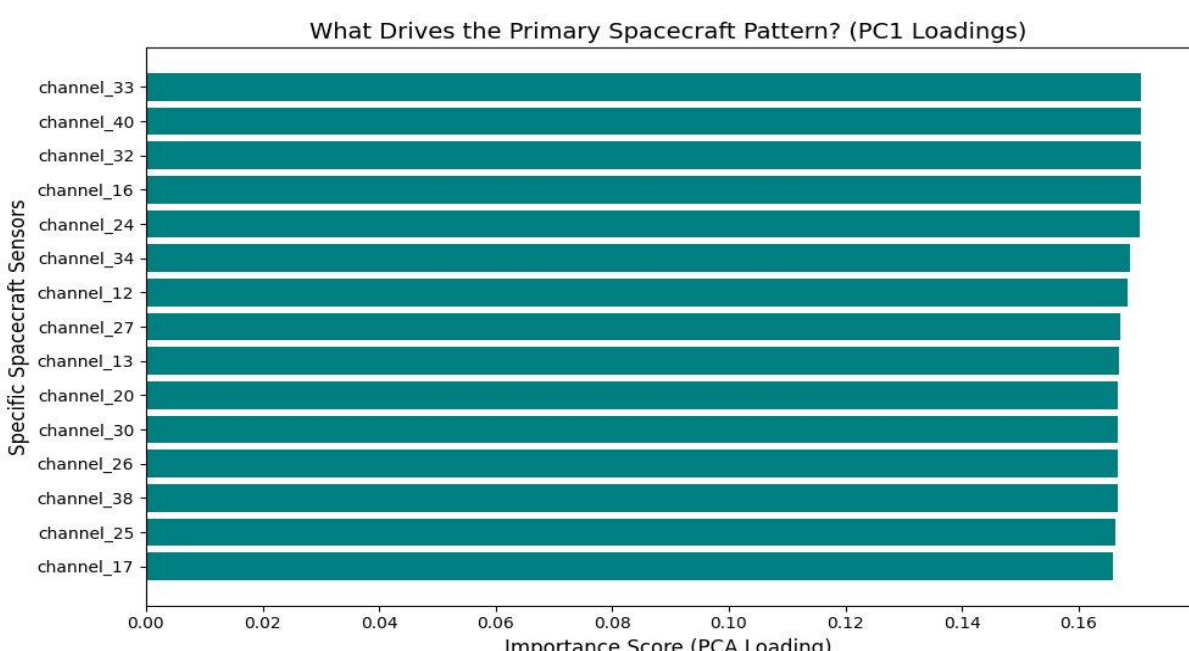


Fig 4. PC1 loadings: near-uniform weights across 15 channels indicate tight thermal/power co-movement. Subsystem-coherent structure recoverable without labels. ✓ *Mission 1 only

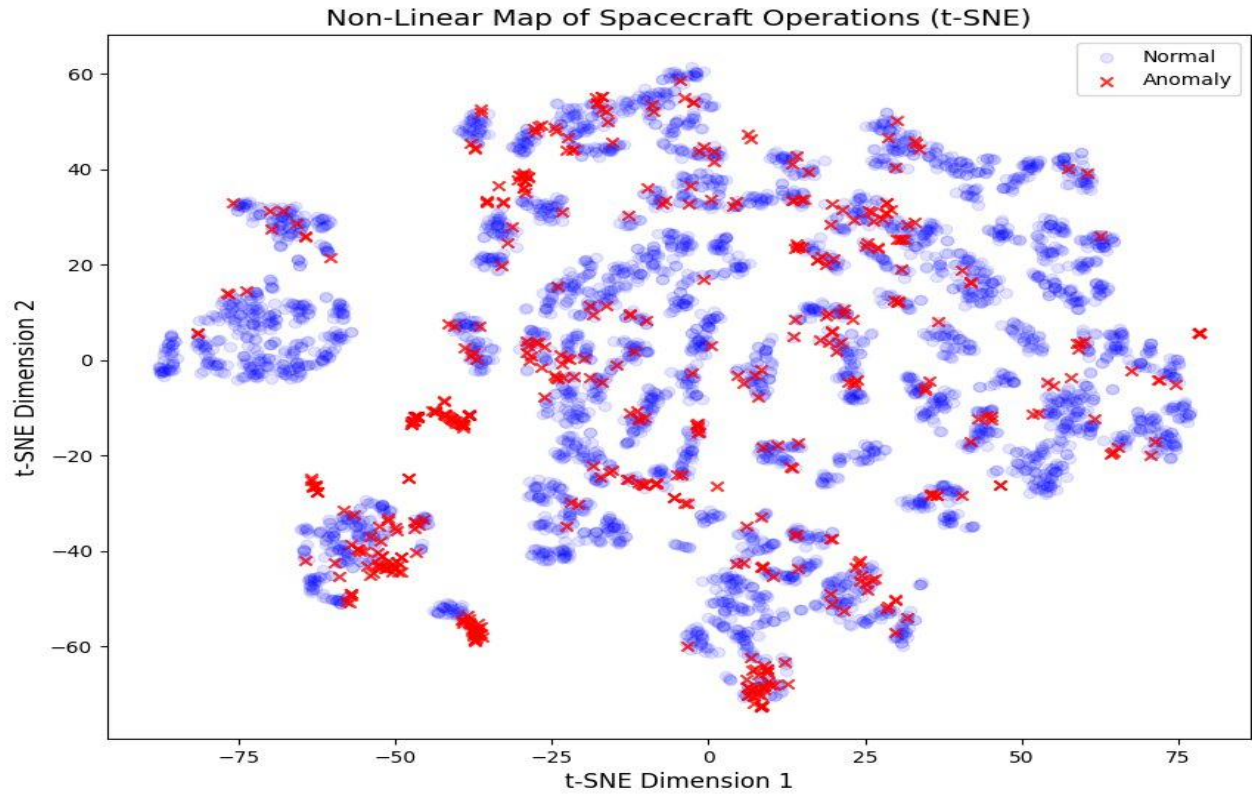


Fig 5. t-SNE of PCA-reduced data: satellite operates in many discrete regimes (fragmented archipelago). Anomalies (red x) scatter across all clusters — no single anomaly zone exists. *Mission 1 only

METHOD 3: NYSTROEM-KPCA

Nystroem approximation to KPCA (RBF kernel) bypasses exact KPCA memory limits on 7.3M samples. 100K training points subsampled as landmarks (n = 150). Grid search: $\text{pca_dim} \in \{5, 10, 15, 25, 40\} \times \gamma \in \{0.1, 0.5, 1.0, 2.0\}/n$. Scored via KPCA feature reconstruction error and centroid distance in kernel space.

KPCA RESULTS & PR-AUC SUMMARY

Nystroem-KPCA regresses below all linear baselines:

KPCA Feat-Recon PR-AUC = **0.251** | KPCA Dist PR-AUC = **0.224**

Curves cross at the optimal $\text{pca_dim} = 25$, signaling high sensitivity to kernel bandwidth γ . High-bandwidth kernels overfit local noise in normal data. Nystroem at 150 landmarks is too sparse to reconstruct the full kernel geometry for a 7.3M-sample dataset. Kernel non-linearity is not the right inductive bias here.

Method	Scoring	PR-AUC	Result
K-Means (raw)	Centroid Dist.	0.322	Baseline
PCA Recon Error	Recon Error	0.322	Neutral =
PCA + K-Means	Centroid Dist.	0.317	Neutral =
KPCA Feat-Recon	Recon Error	0.251	Regress ↓
KPCA Dist	Centroid Dist.	0.224	Regress ↓

Table 1. PR-AUC comparison — all baseline methods on ESA Mission 1 test set.

METHOD 4: GMM PARAMETRIC DISCOVERY

StandardScaler normalization is applied to all channels, followed by PCA compression to 30 components retaining over 99% of Mission 1 variance. A full-covariance GMM is then fit on a 100K reservoir-sampled subset of training data, with K selected by BIC sweep over K=2–30. Reference lines at K=4 (subsystems) and K=18 (groups) let us read off directly whether the model recovers either level of the ESA engineering hierarchy. The anomaly score is the negative log-likelihood of each timestep under the fitted mixture, thresholded at the 99th percentile of training scores.

GMM PARAMETRIC DISCOVERY RESULTS

Method	Mission	BIC K	Subsystems	Groups	ROC-AUC	PR-AUC	F1
GMM (PCA-30)	M1	21	4	18	0.6915	0.3840	0.159
GMM (PCA-30)	M2	29	5	29	0.7639	0.3895	0.307

Table 2. GMM BIC-selected K vs. known ESA subsystem and group counts, with anomaly detection performance on held-out test sets.

METHOD 5: VAE GENERATIVE DISCOVERY

All three VAE architectures operate on sliding windows of W=32 timesteps over StandardScaled channel data with a shared latent dimension of 32. CapacityVAE uses a Conv1D encoder-decoder with capacity-controlled KL annealing; HierarchicalVAE splits the latent space into global (mission-phase) and local (per-channel) components; TransformerVAE replaces the encoder with a Transformer that treats each timestep as a token. All three are scored via reconstruction MSE, negative ELBO, and GMM-OOD, with latent K selected post-hoc by silhouette sweep on encoded training windows.

VAE GENERATIVE DISCOVERY RESULTS

Mission 1					
Method	Score Type	Latent K	ROC-AUC	PR-AUC	F1
CapacityVAE	GMM-OOD	6	0.6280	0.3011	0.157
HierarchicalVAE	GMM-OOD	2	0.5906	0.2932	0.164
TransformerVAE	GMM-OOD	3	0.5904	0.2106	0.082

Table 3. Latent K and anomaly detection performance for three VAE architectures on Mission 1.

Mission 2					
Method	Score Type	Latent K	ROC-AUC	PR-AUC	F1
CapacityVAE	Neg ELBO	16	0.7278	0.4384	0.339
HierarchicalVAE	Recon MSE	28	0.7403	0.4501	0.339
TransformerVAE	Neg ELBO	15	0.7486	0.4518	0.339

Table 4. Latent K and anomaly detection performance for three VAE architectures on Mission 2. Labels are ~95% rare events rather than true anomalies.

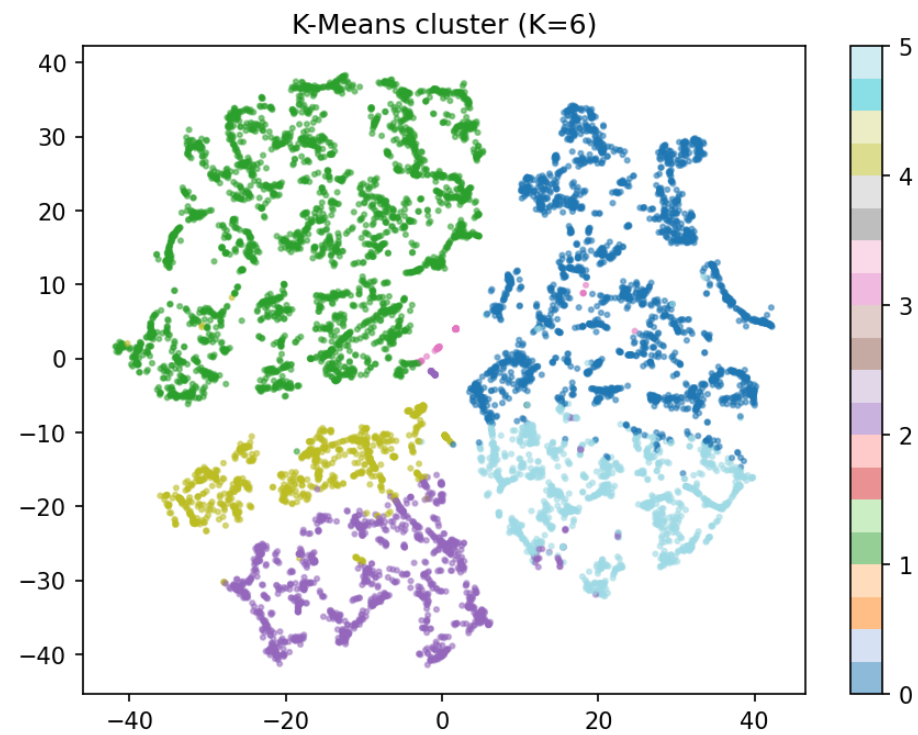


Fig 6. t-SNE of CapacityVAE latent space on **Mission 1**, colored by K-Means cluster (K=6). Six distinct operational modes emerge, sitting between the 4-subsystem and 18-group levels of the ESA engineering hierarchy.

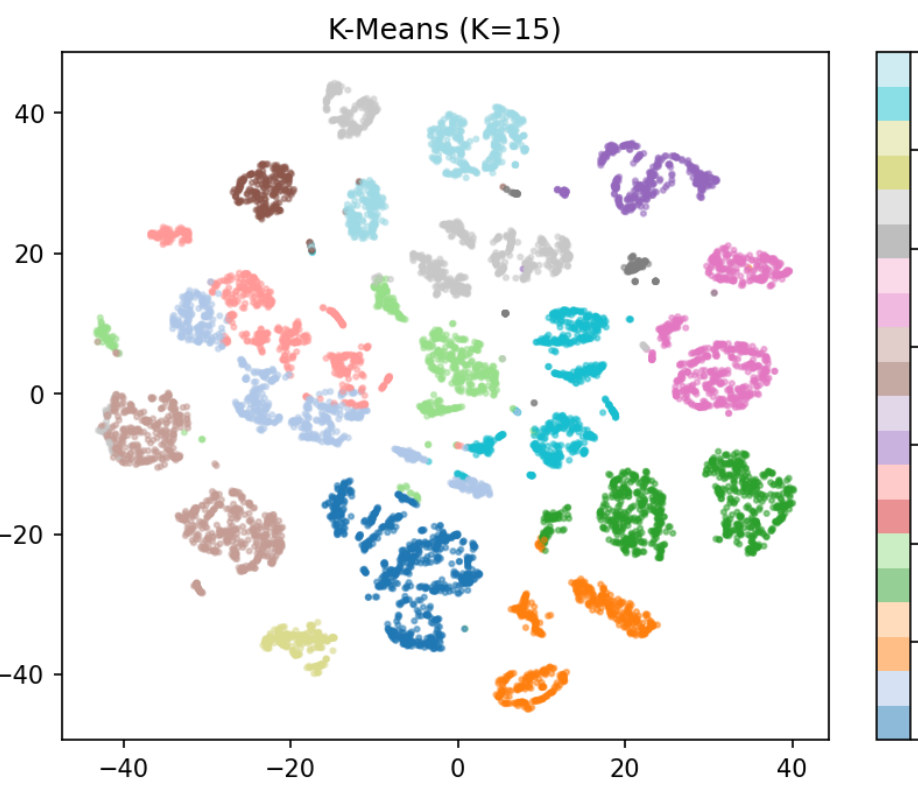


Fig 7. t-SNE of TransformerVAE latent space on **Mission 2**, colored by K-Means cluster (K=15). Fifteen operational modes emerge — above the 5-subsystem level but below the 29-group level of the ESA engineering hierarchy.

DISCUSSION & CONCLUSIONS

- The data has real, compressible latent structure**
33 PCA components retain 95% of variance. PC1 (~37%) has near-uniform loading across a broad channel set — tight thermal/power co-movement. Subsystem-aligned structure recoverable without labels.
- Weakly consistent with the 4-subsystem structure**
BUT ambiguous Silhouette scores are flat across K=2, 3, and 4 (~0.40) before dropping sharply at K=5. The raw 76D space tolerates up to 4 clusters but not more — weakly consistent with Mission 1’s 4 subsystems. However, since K=2, 3, and 4 are indistinguishable by silhouette alone, the raw feature space does not cleanly identify 4 as the natural grouping.
- The latent structure is multi-modal and mode-specific**
t-SNE reveals a fragmented archipelago of operational regimes. Anomalies are scattered across all modes — no global outlier zone exists. Each mode has its own baseline and failure signature.
- Linear and kernel methods cannot exploit this structure**
K-Means, PCA, and KPCA all plateau at PR-AUC ≈ 0.32 or below. KPCA adds non-linearity but hurts (−0.07). The bottleneck is a global model of normality — the data requires mode-local representations (VAE).
- GMM BIC recovers group-level structure, not subsystem-level**
Without any labels, BIC selects K=21 on M1 and K=29 on M2. These sit close to the 18 and 29 engineering group counts — not the 4 and 5 subsystem counts. Unsupervised complexity selection naturally finds fine-grained operational groups, not coarse subsystem boundaries.
- Anomaly detection difficulty depends on anomaly type**
On Mission 1 — where anomalies are true system faults — GMM leads all VAE architectures by a clear margin (0.69 vs. 0.63 ROC-AUC). On Mission 2 — where flagged timesteps are ~95% rare operational events — the gap nearly vanishes (GMM: 0.76, best VAE: 0.75). This suggests VAEs are better suited to detecting distributional novelty while simpler density models hold an edge on fault detection.

FUTURE WORK

- Per-channel anomaly discrimination** — identify which latent dimensions align with known fault channels.
- Latent trajectory over time** — track encoded representations across 84 months to surface operational phase transitions.
- Temporal modeling** — LSTM-VAE or HMM to capture sequential dependencies that window-based VAEs ignore.
- Richer annotation** — incorporating channel metadata and subsystem relationships as clustering constraints to better recover the true engineering hierarchy.